

Being a Giraffe: Embodied Locomotion Interfaces for Understanding Non-Human Morphology in VR

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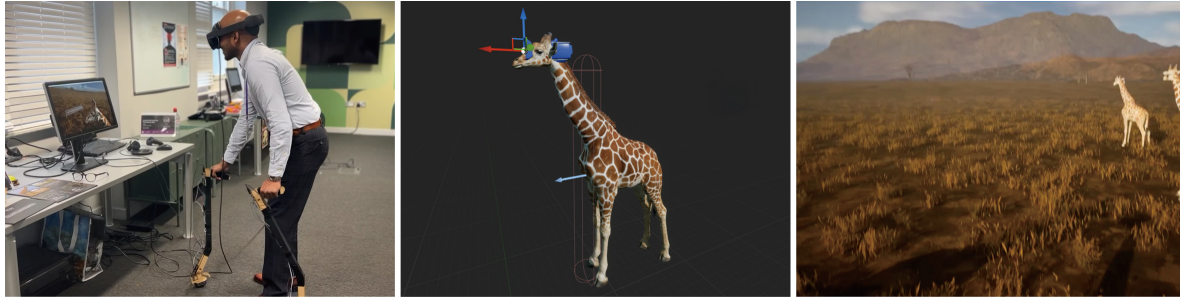


Figure 1: Being a Giraffe system overview with real usage, avatar calibration view and giraffe eye level scene.

ABSTRACT

Can physical embodiment of an animal's body in Virtual Reality (VR) support presence and empathy toward radically different beings? We present *Being a Giraffe*, a VR system that employs custom leg controllers and sustained forward leaning to examine how *productive defamiliarization* shapes embodied animal experiences. Through two-phase exploratory testing, we found that novel physical constraints triggered curiosity and perspective shift, while informational content and atmospheric audio were necessary to maintain presence. We identify preliminary design considerations for balancing physical defamiliarization with multimodal immersion in cross-species embodiment.

Index Terms: Virtual reality—Physical interface design—Animal embodiment—Defamiliarization—Multimodal learning

1 INTRODUCTION

Virtual reality enables perspective-taking through embodiment [1], which can foster empathy and prosocial attitudes [2]. However, most applications maintain familiar interaction paradigms using standard controllers.

When embodying animals with radically different morphology, achieving presence becomes challenging. While previous work has examined observational approaches to wildlife encounters [3] or explored body transfer to morphologically similar species [4], we investigate whether deliberately unfamiliar physical interfaces may support embodied understanding when inhabiting fundamentally different beings.

We selected giraffes due to extreme height (5 meters) and distinctive quadrupedal gait. Contributions include: (1) productive defamiliarization as an emerging design approach, (2) custom locomotion controllers operationalizing this principle, and (3) design insights from two-phase participant testing and reflection on cross-species embodiment.

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2 APPROACH: PRODUCTIVE DEFAMILIARIZATION

We propose *productive defamiliarization* as an emerging design approach for embodied VR that may deconstruct anthropocentric sensorimotor habits by deliberately introducing physical constraints signaling radical bodily difference. We hypothesize it may facilitate perspective-taking through: (1) physical unfamiliarity generating curiosity, (2) novel interaction motivating continued exploration, (3) bodily constraints contextualizing non-human morphology.

3 SYSTEM DESIGN

3.1 Custom Locomotion Controllers

Each custom controller comprises MDF poles, T-bar grips, and 3D-printed PLA hoof bases with dimensions of 715 mm × 170 mm × 60 mm. Spring-loaded hinges allow forward tilt ($\approx 35^\circ$) while hooves maintain ground contact (Figure 2).

Light-dependent resistors (LDR) in each hoof detect steps through light blocking. We calibrated the angle so the shield blocks the LDR only when tilted forward, ensuring reliable detection.

Arduino Uno processes signals and communicates via USB serial to Unreal Engine 5 using the Serial Communication Plugin. Each detected step triggers the giraffe avatar to move forward at approximately 180 cm/s, matching a slow giraffe walking pace.

Users adopt a sustained forward-leaning posture of $\approx 30^\circ$, creating bodily unfamiliarity that breaks default movement schemas and directs attention to the act of locomotion. Our design focuses tracking on forward stepping (via controllers) and head orientation (via HMD), deliberately constraining rather than pursuing full-body mapping to foreground the giraffe's distinctive quadrupedal gait as non-human movement.

3.2 Virtual Environment and Audio Design

The virtual environment uses the 8K Savannah Landscape Pack in Unreal Engine 5, positioning users at giraffe eye-level (5 m). The giraffe avatar is presented from a first-person perspective with visible front legs and shadow when the user looks down, supporting body transfer [5]. Other giraffes and large savanna animals such as elephants and zebras populate the environment.

Thirteen educational text overlays present randomly appearing facts (e.g., "The unique walking gait of giraffes is called 'pacing,' where they move both right legs together, then both left").

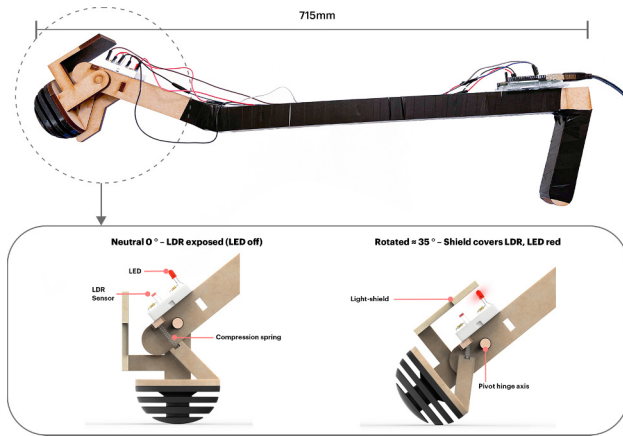


Figure 2: Custom locomotion controller with hoof hinge mechanism and LDR-based tilt detection.

The spatial audio system comprises: (1) synchronized foot-fall sounds, (2) ambient savanna soundscape, and (3) distance-modulated animal vocalizations. Giraffe breathing uses large ruminant sounds at high volume to simulate proximity. Spatial audio allows implicit teaching while text provides explicit knowledge.

4 USER STUDY

4.1 Methodology

We conducted iterative testing with participants in two phases in a 3m × 3m space.

Phase 1 (4 participants): Royal College of Art students experienced controllers and visual environment without audio or text (5-6 minute sessions, Meta Quest 2). All participants provided informed consent. Data collected through semi-structured interviews (n=2) and observation notes (n=4).

Phase 2 (9 participants): Innovate UK Immersive Tech Award attendees experienced the complete system with audio and text (3-5 minute sessions, Meta Quest 2 and Pro). Observations and informal feedback documented during the public showcase.

4.2 Findings

Physical Constraint as Presence Generator: Phase 1 participants (n=4) all reported that the controllers' physical constraint enhanced presence. One Phase 1 participant noted: "The controllers made me think about what being tall with four legs means." (P1)

Reported Connection to Giraffes: Several Phase 2 participants (5 out of 9) felt a closer connection to giraffes, with some spontaneously imagining the giraffe's vulnerability to predators and expressing compassion for its survival.

Multimodal Integration: Phase 1 participants became restless and wanted more context. In Phase 2, participants with audio and text reported stronger immersion: "The footstep sounds made it feel real," (P7) with overlays about gait providing scaffolding.

Affective Response: Participants expressed enjoyment and curiosity, with many noting the novelty of the elevated perspective: "I've never looked down at an elephant before." (P11)

5 DISCUSSION

5.1 Presence Through Physical Constraint

Our preliminary findings suggest productive defamiliarization may offer an alternative to conventional embodiment approaches. Unlike dominant VR paradigms that maximize sensorimotor coherence with human norms, our approach deliberately maintains awareness that "this body moves differently." Participants reported

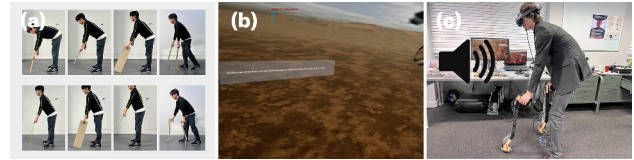


Figure 3: Iterative refinement of productive defamiliarization: (a) exploring body postures and pole configurations; (b) integrating text overlays; (c) complete system with spatial audio.

this physical constraint enhanced their sense of inhabiting another being. Participants' reports align with research showing correlated multisensory cues can support embodiment illusions [6], suggesting this principle could extend to distinctly non-human bodies.

5.2 Orchestrating Multiple Modalities

Our iterative testing revealed that effective animal embodiment may require balancing physical defamiliarization, informational content, and atmospheric audio. Phase 1 observations indicated that physical constraint alone was insufficient without this evolving multimodal support. Figure 3 illustrates this iterative refinement.

5.3 Limits of Cross-Species Embodiment

Humans cannot fully experience non-human consciousness [7], and any VR system inevitably produces human-centered interpretations. Rather than claiming to bridge this gap, productive defamiliarization fosters curiosity and humility toward otherness, a meaningful goal that does not require perfect simulation.

6 CONCLUSION AND FUTURE DIRECTIONS

Being a Giraffe suggests that deliberately unfamiliar physical constraints, combined with atmospheric audio and informational content, may support presence and reflective empathy toward radically different beings. We introduce *productive defamiliarization* as a promising design approach that may deconstruct anthropocentric schemas, with initial evidence suggesting its potential for cross-species embodiment. Future work should explore generalization to other species with distinct morphologies, develop species-specific controllers for educational contexts such as zoos, and conduct more extensive validation studies.

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